

XR-500 Industrial Controller

Installation & Commissioning Guide

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Target Model: XR-500-A1 / XR-500-B1

1. General Overview & Safety Precautions

1.1 Safety Overview

The XR-500 Industrial Controller is a high-reliability control unit designed for heavy manufacturing, process control, and automated fluid management systems. Standard electrical and industrial safety procedures must be observed at all times during installation and commissioning.

- **Qualified Personnel:** Only certified control technicians or licensed electricians should perform electrical wiring and physical mounting.
- **Isolation:** Always verify main incoming power feeds are de-energized, locked out, and tagged out (LOTO) prior to terminal connection.
- **Grounding:** Ensure protective earth (PE) grounding is established with a maximum impedance of $< 0.5 \Omega$.

2. Environmental & Mounting Specifications

2.1 Environmental Requirements

To ensure continuous operation and preserve hardware longevity, the installation location must comply with the environmental operating parameters listed below.

Environmental Parameter	Specification Requirement
Operating Temperature	-10°C to 55°C (14°F to 131°F)
Storage Temperature	-20°C to 70°C (-4°F to 158°F)
Relative Humidity	5% to 90% Non-Condensing

Enclosure Rating	NEMA 4X / IP66
Maximum Vibration	2.5 g RMS (10 Hz – 500 Hz)
Maximum Altitude	2,000 meters (6,560 feet) above sea level

2.2 Mechanical Installation

1. Secure the mounting bracket to a flat, vertical structural surface using four (4) M6 hex bolts torque-rated to 8.5 N·m.
2. Align the mounting tabs on the rear enclosure of the XR-500 with the bracket slots.
3. Slide the unit down until the latch mechanism engages with a distinct click.
4. Verify clear space around the unit housing: minimum 100 mm top/bottom and 50 mm left/right for thermal dissipation.

3. Electrical & Power Wiring

3.1 Power Supply Specifications

The XR-500 contains an auto-sensing internal power supply module designed to accommodate wide input ranges.

- **Primary Power Input:** 24 VDC Nominal (Acceptable Range: 18.0 VDC to 32.0 VDC)
- **Maximum Power Consumption:** 45 W Peak (32 W typical)
- **Inrush Current:** < 12 A for < 5 ms
- **Circuit Breaker Requirement:** 5 A slow-blow fuse or dedicated miniature circuit breaker (MCB)

3.2 Terminal Pinouts

Connect all primary power wiring to Terminal Block A (TB-A) using 16 AWG stranded copper wire rated for 90°C.

- **TB-A Pin 1:** +24 VDC Line
- **TB-A Pin 2:** DC Return (GND)
- **TB-A Pin 3:** Protective Earth (PE Chassis Ground)

4. Initial Commissioning Procedure

4.1 Pre-Power Checks

1. Inspect all physical wire connections for correct polarity and mechanical firmness.
2. Confirm that analog and digital expansion module ribbon cables are fully seated.
3. Verify line pressure on connected fluid lines is reduced to ambient prior to initial power-up.

4.2 Initial Power-Up Sequence

1. Apply 24 VDC power to TB-A.
2. Observe the front panel diagnostic LEDs:
 - **POWER LED:** Solid Green within 3 seconds.
 - **STATUS LED:** Blinking Amber during system bootloader execution (approx. 20 seconds).
 - **SYSTEM READY LED:** Solid Green upon boot completion.
3. Connect an Ethernet cable from the configuration workstation to Port 1 (ETH1).
4. Default Factory IP Address: 192.168.1.100 (Subnet Mask: 255.255.255.0).
5. Open a browser and navigate to [\[http://192.168.1.100\]](http://192.168.1.100)(<http://192.168.1.100>) to launch the Initial Setup Wizard.